



ORBITGUARD-Space Debris Monitoring and Collision Risk Assessment Platform

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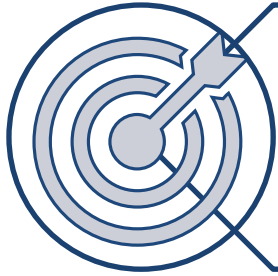
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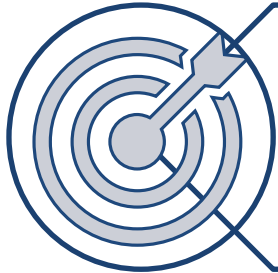
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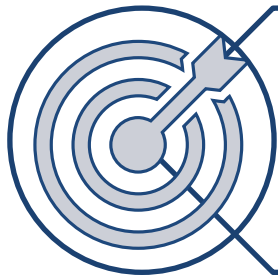
Introduction...



Overview: Develops a real-time platform to visualize satellites and space debris in a 3D environment while monitoring their orbital movements and collision risks..



Problem Statement: The growing amount of space debris increases the risk of collisions with operational satellites, requiring an efficient real-time monitoring and risk assessment system.



Purpose: To analyze live TLE data, track orbital objects, assess potential collision risks, and provide an interactive dashboard for improved space situational awareness..

Introduction



Space debris and active satellites continuously orbit the Earth, making space traffic increasingly complex. **OrbitGuard** is a real-time 3D visualization and collision risk assessment platform that tracks orbital objects using live TLE data, helping users monitor satellite movements and identify potential collision threats through an interactive mission control dashboard.



Objectives

Collect and process live **Two-Line Element (TLE)** data from CelesTrak to track active satellites and space debris in real time.

Calculate orbital positions, velocities, and trajectories using the **Skyfield (SGP4)** propagation model for accurate visualization.

Assess collision risks by measuring the proximity between operational satellites and space debris using distance-based algorithms.

Provide real-time alerts and insights to improve space situational awareness and support safer satellite operations.

Develop an interactive **3D mission control dashboard** to visualize orbital objects, telemetry data, and collision risk levels.



Engineer to Excel

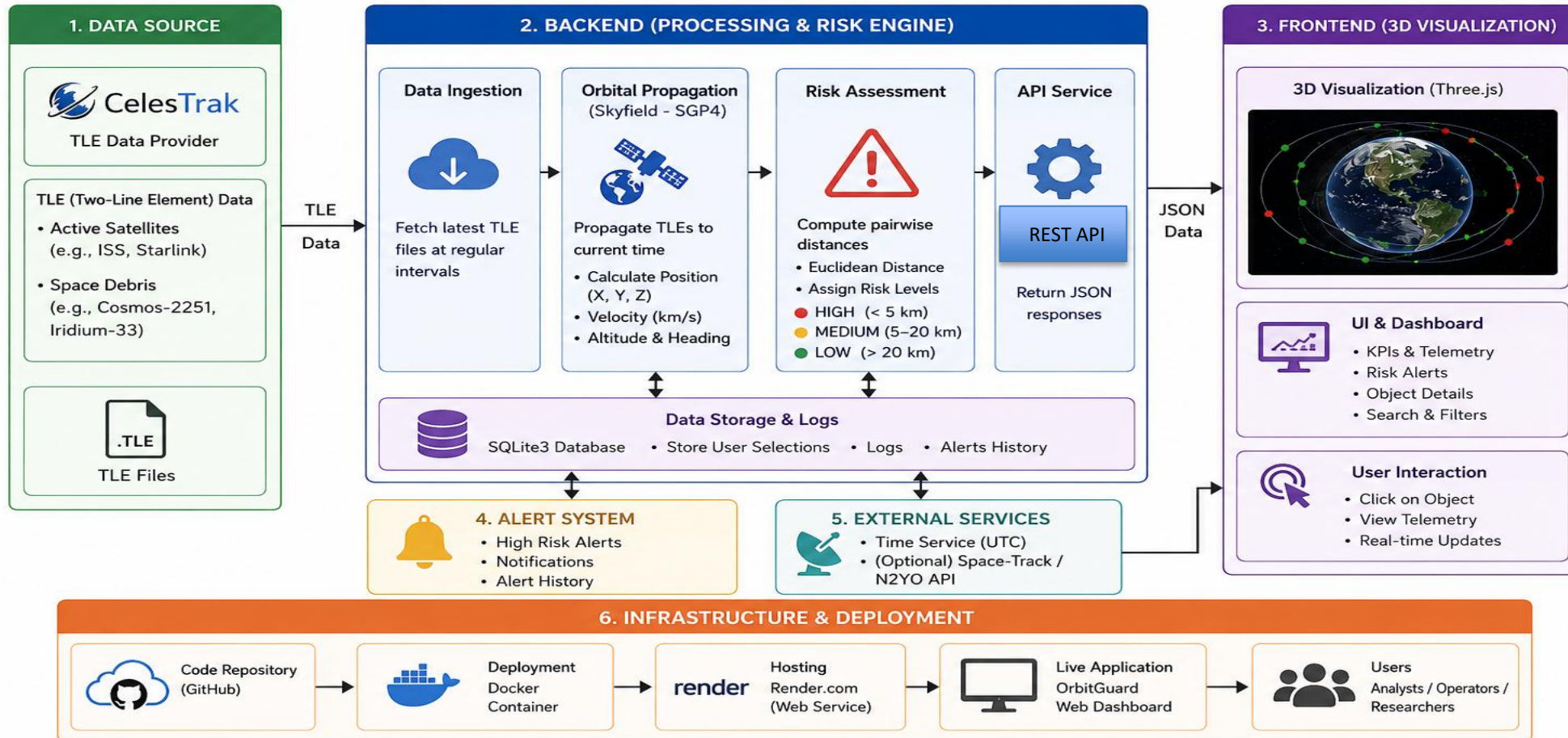
TOOLS AND TECHNOLOGIES



Category	Tools/Technologies	Purpose
Programming	JAVA, React js	Backend processing and frontend functionality
Web Framework	SPRINGBOOT	Build and serve high-performance REST APIs
Data Source	CelesTrak (TLE Data)	Provides real-time satellite and space debris data
Database	SQLite3	Storing and retrieving past user-selected views
Orbital Processing	Skyfield, SGP4	Calculate satellite positions, velocity, and trajectories
User Interface	Bootstrap (via external CSS)	Design the mission control dashboard
3D Visualization	Three.js (WebGL)	Render Earth, satellites, and debris in 3D

Methodology

ORBITGUARD – ARCHITECTURE DIAGRAM



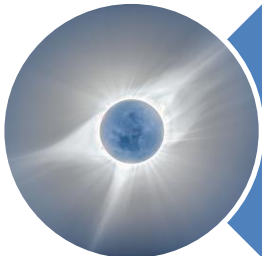
IMPLEMENTATION



Initially, **live TLE data** of active satellites and space debris was collected from **CelesTrak**, processed using the **Skyfield (SGP4)** propagation model to determine their real-time orbital positions and trajectories.



An interactive **3D mission control dashboard** was then developed using **Three.js**, and **JavaScript** to visualize satellites, monitor telemetry, and dynamically assess collision risks between operational satellites and nearby space debris.

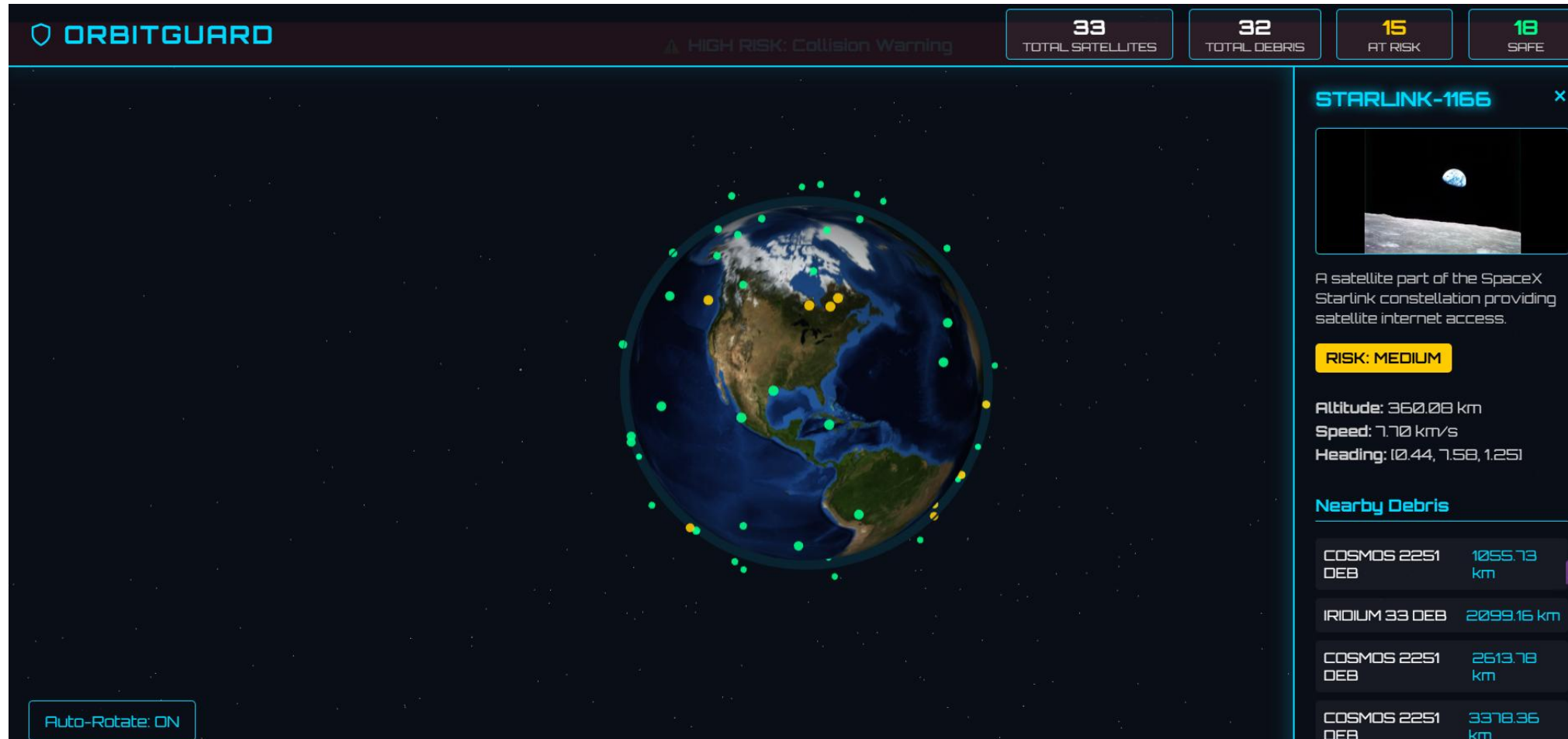


To provide a clear overview of the current orbital environment, enabling users to monitor active satellites, track space debris, and identify high-risk objects for improved space situational awareness.



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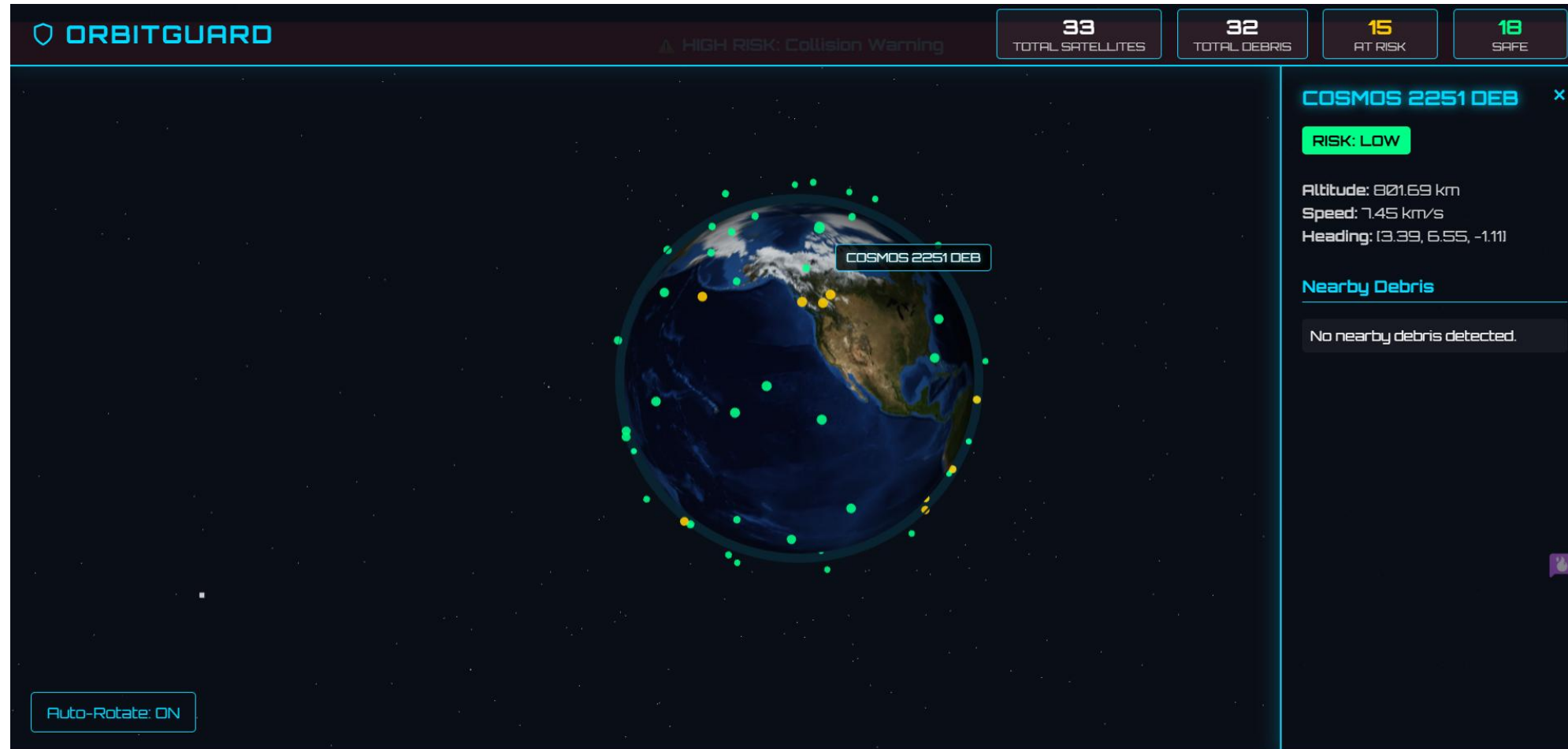
Implementation....





Engineer to Excel

Implementation....





Engineer to Excel

RESULT AND DISCUSSION...



STARLINK-1080 ×



A satellite part of the SpaceX Starlink constellation providing satellite internet access.

RISK: LOW

Altitude: 351.37 km
Speed: 7.70 km/s
Heading: [-4.66, 4.18, -4.49]

Nearby Debris

COSMOS 2251 DEB	2682.48 km
COSMOS 2251 DEB	2781.71 km
IRIDIUM 33 DEB	2916.3 km
COSMOS 2251 DEB	3991.78 km

SATELLITES INFO

COSMOS 2251 DEB ×

RISK: LOW

Altitude: 801.69 km
Speed: 7.45 km/s
Heading: [3.39, 6.55, -1.11]

Nearby Debris

No nearby debris detected.

DEBRIS INFO

33 TOTAL SATELLITES	32 TOTAL DEBRIS	15 AT RISK	18 SAFE
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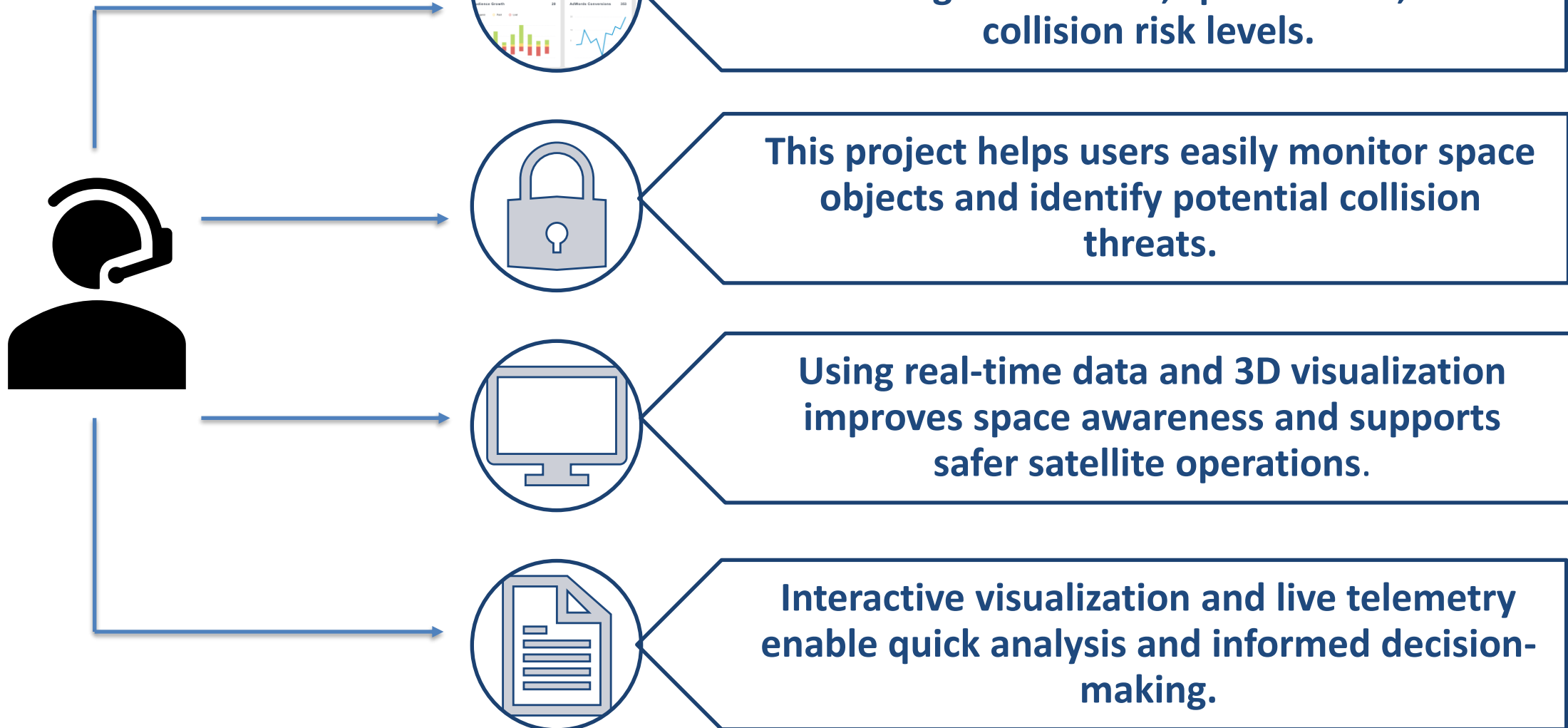
KPI

RESULT AND DISCUSSION

PROTOTYPE LINK : <https://orbitguard-wrz6.onrender.com>



Conclusion



References

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